IOT-BASED LABORATORY EQUIPMENT TRACKING FOR COMPUTING LABORATORIES

A Capstone Project
Presented to the Faculty of the
School of Computing
Holy Angel University

In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Computer Science

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Proposal manuscript — draft (Introduction to Conceptual Framework). Later sections follow in the complete proposal. Dummy document generated for testing the HAU-SOC Thesis & Capstone prototype. All names, data and references are placeholders.

APPROVAL SHEET

This capstone entitled **"IOT-BASED LABORATORY EQUIPMENT TRACKING FOR COMPUTING LABORATORIES"**, prepared and submitted in partial fulfillment of the requirements for the degree Bachelor of Science in Computer Science has been examined and is recommended for acceptance and oral examination.

Dean Delta
Adviser

ORAL EXAMINATION

Approved by the committee of oral examiners on
[Date of defense]

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Panel Chair

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APPROVAL

Accepted and approved in partial fulfillment of the requirements for the degree Bachelor of Science in Computer Science.

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Acknowledgment

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IoT-Based Laboratory Equipment Tracking for Computing Laboratories

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Abstract

The abstract is written after the study is completed and is left blank at the proposal stage.

IOT-BASED LABORATORY EQUIPMENT TRACKING FOR COMPUTING LABORATORIES

Introduction

The computing laboratories lend projectors, development boards, networking kits and spare peripherals to classes every day. Borrowing is recorded on paper, items are moved between rooms without a record, and the semestral inventory regularly finds equipment missing or in the wrong room.

Bluetooth Low Energy (BLE) tags are inexpensive and last months on a coin cell. Gateways in each room can detect the tags and report their signal strength, from which a server can estimate the room an item is in. Combined with a borrowing log, such a system would show where every item is and flag items that leave the laboratories.

The problem this study addresses is the following: laboratory staff cannot locate equipment without searching every room, borrowing records are incomplete, and losses are discovered only at inventory time. This capstone project therefore proposes the equipment tracker.

Review of Related Literature

This section reviews studies and existing systems related to the equipment tracker and the methods used to evaluate it.

Indoor Positioning with Bluetooth Low Energy

RSSI-based room-level positioning with one receiver per room achieves over 90% accuracy when readings are smoothed over a short window (Placeholder & Author, 2023).

Asset Tracking in Educational Laboratories

Tagged asset systems in school laboratories reduced search time and made borrowing records complete (Sample, 2024).

Signal Smoothing and Classification

Moving averages and Kalman filters reduce RSSI noise; a nearest-gateway rule with hysteresis prevents items from flickering between rooms (Example & Writer, 2022).

Evaluating IoT Systems

ISO/IEC 25010 is commonly applied to IoT prototypes alongside accuracy measurements of the sensing layer (Example, 2023).

Synthesis

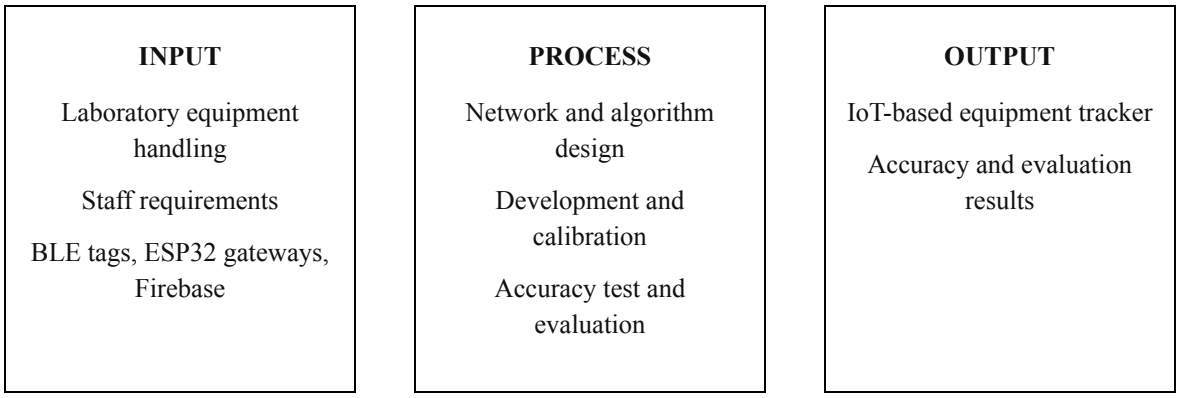
The reviewed studies show that the problem is common and that a focused, web-based solution can address it at low cost. None of the systems reviewed fits the setting of this study exactly, which is the gap the equipment tracker fills.

Conceptual Framework

The study follows the Input–Process–Output model shown in Figure 1.

Figure 1

Conceptual Framework of the Study



Draft: Objectives of the Study, Scope and Delimitations and Method are being written and will be in the complete proposal manuscript.